

MANAGEMENT INFORMATION SYSTEM

Unit-2: Data Processing & Information System for Decision Making

Comprehensive Study Notes

PART A: DATA PROCESSING

1. OPERATION OF MANUAL INFORMATION SYSTEM

1.1 Introduction to Manual Information Systems

Definition: A Manual Information System is a system where information is collected, processed, stored, and distributed using manual methods without the use of computers or automated technology.

Characteristics of Manual Information Systems:

Characteristic	Description
Paper-Based	Information recorded on paper forms, ledgers, files
Human-Intensive	Requires significant human effort for processing
Manual Calculations	Arithmetic done by hand or using calculators
Physical Storage	Files, cabinets, rooms for storing documents
Slow Processing	Time-consuming data handling
Prone to Errors	Manual data entry leads to mistakes
Limited Access	Only those with physical access can view

1.2 Components of Manual Information System

[Manual Information System Components Flowchart: People → Procedures → Data/Records → Storage → Output]

1.3 Operation Flow in Manual Information System

[Operation Flow: Data Collection → Data Recording → Data Processing → Data Storage → Reporting]

1.4 Types of Manual Information Systems

Type	Description	Example
Manual Accounting System	Records all financial transactions manually	Cash book, ledger, journal
Manual Inventory System	Tracks stock levels using physical counts	Stock register, bin cards
Manual Personnel System	Maintains employee records	Employee files, attendance register
Manual Sales System	Records sales transactions	Sales register, order forms
Manual Purchase System	Tracks procurement	Purchase orders, supplier files
Manual Reporting System	Prepares manual reports	Monthly sales reports, stock reports

1.5 Advantages and Disadvantages of Manual Systems

Advantages:

Advantage	Description
Low Cost	No expensive hardware/software needed
Simple	Easy to understand and operate
No Power Dependency	Works without electricity
Tangible Records	Physical documents are reassuring
Easy to Change	Changes are simple to implement
No Technical Skills	Requires basic literacy only

Disadvantages:

Disadvantage	Description
Slow Processing	Takes time to process large volumes
Prone to Errors	Manual calculations and entries cause errors
Limited Capacity	Can only handle limited data volume
Storage Issues	Requires large physical storage space
Security Problems	Documents can be lost or stolen
Duplication	Same data entered multiple times
No Backup	Difficult to backup physical records
Limited Analysis	Cannot perform complex analysis
No Sharing	Information sharing is difficult
Time-Consuming	Many hours spent on routine tasks

2. COMPUTERIZED INFORMATION SYSTEM

Definition: A Computerized Information System is a system that uses computer hardware and software to collect, process, store, and distribute information efficiently and accurately.

Advantages over Manual Systems:

- **Speed:** Processes data in seconds/minutes vs hours/days
- **Accuracy:** 99.9% accuracy vs 80-90% in manual
- **Storage:** Massive data in minimal space
- **Retrieval:** Instant search and access
- **Sharing:** Multiple users can access simultaneously
- **Analysis:** Complex analysis and reporting possible
- **Backup:** Easy data backup and recovery
- **Integration:** Data flows across departments

2.1 Types of Data Processing

A. Batch Processing

Data is collected, stored, and processed in groups (batches) at scheduled intervals.

- **Examples:** Payroll processing, billing, bank statement generation
- **Advantages:** Efficient for large volumes, lower cost
- **Disadvantages:** Delayed results, no real-time updates

B. Online / Real-Time Processing

Data is processed immediately as it is entered into the system.

- **Examples:** ATM transactions, online booking, POS systems
- **Advantages:** Immediate results, up-to-date information
- **Disadvantages:** Higher cost, requires continuous connectivity

C. Time-Sharing Processing

Multiple users share computer resources simultaneously through rapid switching.

- **Examples:** Mainframe with multiple terminals
- **Advantages:** Resource sharing, cost-effective
- **Disadvantages:** Performance degrades with many users

3. COMPONENTS OF COMPUTER SYSTEM

3.1 Introduction

Definition: A computer system is a combination of hardware and software components that work together to process data and produce information.

[Computer System Components: Hardware + Software + Data + People + Procedures]

3.2 Hardware Components

A. Input Devices

Device	Function	Example Use
Keyboard	Text input	Data entry, commands
Mouse	Pointing and selection	Navigation, selection
Scanner	Convert paper to digital	Document scanning
Barcode Reader	Read barcodes	Product scanning
Touch Screen	Touch-based input	Mobile devices, kiosks
Microphone	Audio input	Voice commands
Biometric Devices	Fingerprint, iris scan	Security systems

B. Central Processing Unit (CPU)

Definition: The CPU is the "brain" of the computer that executes program instructions.

[CPU Components: ALU + Control Unit + Registers]

Component	Function
ALU (Arithmetic Logic Unit)	Performs arithmetic and logical operations
Control Unit	Directs system operations, fetches and executes instructions
Registers	Small, fast memory for temporary data storage

C. Memory

Type	Description	Characteristics
RAM (Random Access Memory)	Main memory, volatile	Fast, lost when power off
ROM (Read Only Memory)	Permanent memory	Non-volatile, cannot be changed
Cache Memory	Very fast, small memory	Between CPU and RAM
Secondary Storage	Permanent storage	Slower, non-volatile

[Memory Hierarchy: Registers → Cache → RAM → Secondary Storage]

D. Storage Devices

Device	Capacity	Speed	Use
Hard Disk Drive (HDD)	High	Medium	General storage
Solid State Drive (SSD)	Medium	Fast	Fast access storage
USB Flash Drive	Low-Medium	Medium	Portable storage
CD/DVD	Medium	Slow	Archive, distribution
Cloud Storage	Variable	Depends on internet	Remote access, backup

E. Output Devices

Device	Function	Example
Monitor	Visual output	Displaying information
Printer	Hard copy output	Printing reports
Speaker	Audio output	Sound alerts
Projector	Large display	Presentations

3.3 Software Components

A. System Software

Type	Description	Examples
Operating System	Manages hardware and software	Windows, Linux, macOS
Device Drivers	Communicate with hardware	Printer drivers, graphics drivers
Utility Programs	System maintenance	Antivirus, backup software
Programming Languages	Create software	C, Java, Python

B. Application Software

Type	Description	Examples
Business Software	Business operations	MIS, CRM, ERP
Office Suite	Office productivity	MS Office, Google Workspace
Database Software	Data management	MySQL, Oracle
Communication	Communication tools	Email, Slack
Industry-Specific	Custom applications	Hospital management, Banking

3.4 Data Component

Definition: Data refers to raw facts and figures that are processed by the computer system.

Type	Description	Example
Structured Data	Organized, formatted	Database tables, spreadsheets
Unstructured Data	No specific format	Emails, videos, documents
Semi-Structured Data	Some organization	XML, JSON

3.5 People & Procedures

People (User Types)	Role
End Users	Use the system for their work
System Analysts	Design and develop systems
Programmers	Write and maintain software
Database Administrators	Manage databases
Network Administrators	Manage networks
IT Managers	Manage IT resources

Procedures	Description
Operating Procedures	How to use the system
Security Procedures	Protecting data and systems
Backup Procedures	Data backup and recovery
Maintenance Procedures	System maintenance
Disaster Recovery	Restoring operations after failure

4. CONVERSION OF MANUAL TO COMPUTER-BASED SYSTEMS

4.1 Introduction

Definition: Conversion from manual to computer-based systems is the process of replacing manual information processing methods with computer-based systems to improve efficiency, accuracy, and effectiveness.

4.2 Need for Conversion

Reason	Description
Information Overload	Manual systems can't handle growing data
Accuracy Issues	Manual errors are unacceptable
Time Constraints	Need faster processing
Competitive Pressure	Need to keep up with competitors
Information Needs	Complex analysis required
Cost Considerations	Long-term cost savings
Regulatory Requirements	Compliance needs
Global Operations	Need to work across locations

4.3 Steps in Conversion Process

[Conversion Process: Need Assessment → Feasibility Study → System Planning → System Design → Development → Testing → Data Migration → Training →

Implementation → Evaluation → Maintenance]

1. Need Assessment

- Identify problems with current manual system
- Define information requirements
- Establish goals for automation

2. Feasibility Study

Feasibility Type	Questions to Ask
Technical	Is technology available? Do we have skills?
Economic	Is it cost-effective? What is ROI?
Operational	Will users accept it? Will it work?
Schedule	Can we implement on time?
Legal	Are there legal constraints?

3. System Planning

- Define project scope
- Set timeline and budget
- Allocate resources
- Create project team

4. System Design

- Define system requirements
- Design database
- Design user interface
- Design reports and outputs

5. Development

- Write software

- Configure hardware
- Set up networks
- Create documentation

6. Testing

Test Type	Description
Unit Testing	Test individual components
Integration Testing	Test combined components
System Testing	Test entire system
User Acceptance Testing	Users test the system

7. Data Migration

- Clean existing data
- Convert data format
- Transfer to new system
- Verify data integrity

8. Training

Training Type	Audience
User Training	End users
Technical Training	IT staff
Management Training	Managers

9. Implementation Strategies

Strategy	Description	Advantages	Disadvantages
Direct Cutover	Old system stopped, new started	Fast, less cost	High risk
Parallel Conversion	Both systems run simultaneously	Low risk, easy to verify	Costly, time-consuming

Phased Conversion	Implement in stages	Manageable risk	Takes long
Pilot Conversion	Implement in one dept first	Learning, refinement	Slow rollout

10. Evaluation & Maintenance

- Check if system meets objectives
- Measure performance improvements
- Gather user feedback
- Fix bugs and add improvements

4.4 Challenges in Conversion

Challenge	Description	Solution
Employee Resistance	Fear of change	Training, involvement
Data Migration Issues	Data loss, errors	Careful planning, testing
Cost Overruns	Higher than budget	Realistic planning
Time Delays	Longer than planned	Realistic schedules
Technical Problems	Compatibility issues	Thorough testing
Management Support	Lack of commitment	Executive sponsorship

5. TRANSACTION PROCESSING SYSTEM (TPS)

Definition: A Transaction Processing System (TPS) is a computerized system that processes routine business transactions efficiently and accurately.

Features of TPS:

- **High Volume:** Processes large number of transactions
- **Routine:** Handles standard, repetitive tasks
- **Accuracy Critical:** Errors have major impact
- **Real-time:** Often needs immediate processing

Types of TPS:

Type	Description	Example
Batch Processing TPS	Process transactions in groups at scheduled times	Payroll, billing, bank statement generation
Real-Time TPS	Process transactions immediately as they occur	ATM, online orders, POS systems

Examples of TPS:

- **Order Processing:** Order entry, verification, inventory update, invoicing
- **Payroll Processing:** Salary calculation, deductions, payslip generation
- **Accounting System:** Accounts payable/receivable, general ledger

6. DATA BANK CONCEPT

Definition: A Data Bank is a centralized repository of data that is organized, stored, and made available for use by various applications and users within an organization.

Data Bank vs Database:

Aspect	Data Bank	Database
Scope	Broader, includes data + management	Specifically data storage
Purpose	Information resource	Data management
Content	Data, metadata, data dictionaries	Data organized in tables
Access	Multiple applications, users	Applications, users
Management	Data administration	Database administration

Benefits of Data Bank:

- **Data Integration:** All data in one place
- **Reduced Redundancy:** Avoids duplicate data
- **Data Consistency:** Same data used by all
- **Better Access:** Easy to find and retrieve
- **Data Security:** Centralized security
- **Data Quality:** Better data quality control
- **Flexibility:** Adaptable to new needs
- **Cost Efficiency:** Less duplication of effort

Role of Data Bank in MIS:

Function	Description
Data Source	Provides data for information generation

Historical Record	Maintains organizational history
Decision Support	Provides data for decisions
Performance Monitoring	Tracks organizational performance
Trend Analysis	Enables identification of patterns
Compliance	Supports regulatory reporting

PART B: INFORMATION SYSTEM FOR DECISION MAKING

7. INFORMATION SYSTEM FOR DECISION MAKING

7.1 Introduction to Decision Making

Definition: Decision making is the process of selecting the best course of action from available alternatives to achieve organizational goals.

7.2 Types of Decisions

[Decision Types: Structured (Programmed) vs Unstructured (Non-Programmed)]

A. Programmed (Structured) Decisions

Aspect	Description
Nature	Routine, repetitive
Procedure	Standard rules exist
Frequency	High
Information	Well-defined
Level	Usually lower management
Example	Reorder point, approving leave

B. Non-Programmed (Unstructured) Decisions

Aspect	Description
Nature	Novel, unique

Procedure	No standard procedure
Frequency	Low
Information	Ambiguous, incomplete
Level	Usually higher management
Example	New investment, market entry

7.3 Decision Making Process

[7-Step Decision Making Process: Identify Problem → Gather Information → Develop Alternatives → Evaluate Alternatives → Choose Best → Implement → Evaluate]

Step	Description	MIS Role
1. Identify Problem	Recognize that a decision is needed	Alert systems, exception reports
2. Gather Information	Collect relevant data	Access data and reports
3. Develop Alternatives	Identify possible solutions	Scenario generation
4. Evaluate Alternatives	Assess pros and cons	Analysis, modeling
5. Choose Best Alternative	Select optimal solution	Decision support tools
6. Implement Decision	Put decision into action	Implementation planning
7. Evaluate Decision	Check if decision worked	Feedback, monitoring

7.4 Decision Making Levels

[Decision Levels: Strategic (Top Mgmt) → Tactical (Middle Mgmt) → Operational (Lower Mgmt)]

Level	Decision Type	Information Source	MIS Support
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Strategic	Non-programmed	External, internal	EIS, BI
Tactical	Semi-programmed	Internal, some external	DSS, MIS
Operational	Programmed	Internal	TPS, MIS reports

Information Needs at Different Levels:

Level	Information Need	Characteristics
Operational	Detailed, current, internal	High volume, low summary
Tactical	Summarized, historical, internal	Medium volume, some summary
Strategic	Aggregated, future-oriented, external	Low volume, high summary

8. DECISION SUPPORT SYSTEM (DSS)

Definition: A Decision Support System (DSS) is a computer-based information system that supports business or organizational decision-making activities by providing analytical tools and models to help managers solve semi-structured and unstructured problems.

8.1 Components of DSS

[DSS Components: Database → Model Base → User Interface]

Component	Description
Database	Stores data from internal and external sources for decision-making
Model Base	Collection of mathematical and analytical models for data analysis
User Interface	Allows users to interact with the DSS, input parameters, view results

8.2 Types of DSS

Type	Description	Example
Data-Driven DSS	Focuses on access to and analysis of large data sets	OLAP, data mining, dashboards
Model-Driven DSS	Uses mathematical and statistical models for analysis	Financial modeling, simulation
Knowledge-Driven DSS	Provides specialized problem-solving expertise	Expert systems, recommendation systems

8.3 DSS Applications

- **What-if Analysis:** Changing variables to see impact on outcomes
- **Scenario Planning:** Evaluating alternative future scenarios
- **Financial Modeling:** Investment analysis, budgeting
- **Risk Analysis:** Assessing and quantifying risks

9. EXECUTIVE INFORMATION SYSTEM (EIS)

Definition: An Executive Information System (EIS) is a computer-based system designed specifically for senior executives to access summarized, relevant information about organizational performance and external conditions to support strategic decision-making.

Features of EIS:

- **Dashboard:** Visual display of key metrics and KPIs
- **Drill-down:** Ability to view detailed data on request
- **External Data:** Market trends, competitor information
- **Easy to Use:** Minimal training required, intuitive interface
- **Strategic Focus:** Supports long-term strategic decisions
- **Summarized View:** High-level aggregated information

10. EXPERT SYSTEMS

Definition: An Expert System is a computer program that uses artificial intelligence techniques to mimic the decision-making ability of a human expert in a specific domain.

Components of Expert Systems:

Component	Description
Knowledge Base	Contains domain-specific facts and rules
Inference Engine	Applies rules to facts to derive conclusions
User Interface	Allows users to query and receive explanations

Applications:

- Medical diagnosis
- Fault detection
- Financial planning
- Customer support

11. EVOLUTION OF INFORMATION SYSTEM

[IS Evolution Timeline: EDP (1950s) → MIS (1960s) → DSS (1970s) → EIS/Strategic (1980s) → Internet (1990s) → Social/Cloud (2000s) → AI/Analytics (2010s+)]

Summary of Evolution:

Decade	Era	Focus	Technology	Key Concept
1950s-60s	EDP	Automation	Mainframes	Data Processing
1960s-70s	MIS	Reporting	Mainframes+	Information
1970s-80s	DSS	Decision Support	Minis/PCs	Decision Making
1980s-90s	Strategic	Advantage	PCs/Networks	Strategy
1990s-00s	Internet	Connectivity	Internet	Integration
2000s-10s	Social/Cloud	Collaboration	Cloud/Mobile	Accessibility
2010s+	AI/Analytics	Intelligence	AI/ML	Insights

12. TYPES OF COMPUTER-BASED APPLICATIONS

Comparison of Application Types:

Application	Users	Purpose	Type of Data	Example
TPS	Operational staff	Transaction processing	Operational	Sales orders
MIS	Middle managers	Reporting	Summarized	Sales reports
DSS	Analysts, managers	Decision support	Various	What-if analysis
EIS	Executives	Strategic decisions	Summarized, external	Dashboard
ERP	All departments	Integration	Cross-functional	SAP, Oracle
CRM	Sales, Marketing	Customer management	Customer data	Salesforce

[Application Hierarchy: TPS (Bottom) → MIS → DSS → EIS (Top)]

KEY DIAGRAMS FOR EXAM

Diagram	Description
1. Manual System Flow	Data Collection → Recording → Processing → Storage → Reporting
2. Computer System Components	Hardware, Software, Data, People, Procedures
3. CPU Architecture	ALU, CU, Registers
4. Memory Hierarchy	Registers → Cache → RAM → Secondary Storage
5. Conversion Steps	Need Assessment → Feasibility → Design → etc.
6. Implementation Strategies	Direct, Parallel, Phased, Pilot
7. Application Types	TPS, MIS, DSS, EIS, ERP, CRM
8. Decision Making Process	7-step model
9. Decision Types	Programmed vs Non-programmed
10. Decision Levels	Operational, Tactical, Strategic
11. IS Evolution Timeline	EDP → MIS → DSS → EIS → Internet → Cloud → AI

IMPORTANT DEFINITIONS

Term	Definition
Manual Information System	Paper-based system with manual processing
Computer System	Hardware, software, data, people, procedures
CPU	Central Processing Unit - processor
Data Bank	Centralized data repository
TPS	Transaction Processing System - processes routine transactions
DSS	Decision Support System - supports complex decisions
EIS	Executive Information System - for senior executives
ERP	Enterprise Resource Planning - integrated business systems
Decision Making	Selecting best course of action
Programmed Decision	Routine, structured decision
Non-Programmed Decision	Novel, unstructured decision
Bounded Rationality	Limited by information/time/ability

PRACTICE QUESTIONS

Long Questions (10 marks)

1. Explain the operation of manual information systems with advantages and disadvantages.
2. Describe the components of a computer system with a neat diagram.
3. Explain the process of converting manual systems to computer-based systems.
4. What is a data bank? Explain its concept and benefits.
5. Discuss various types of computer-based applications.
6. Explain the evolution of information systems with appropriate diagrams.
7. Describe the decision-making process and the role of MIS in decision making.
8. Explain Decision Support System (DSS) with its components and types.

Short Questions (5 marks)

1. Discuss the advantages and disadvantages of manual systems.
2. Explain the components of a computer system.
3. What is the difference between data, information, and knowledge?
4. What is a data bank? Why is it important?
5. Explain TPS and its characteristics.
6. Compare MIS and DSS.
7. What is bounded rationality in decision making?
8. Explain any two implementation strategies.
9. Describe the role of MIS in decision making.

10. Differentiate between programmed and non-programmed decisions.

Very Short Questions (2 marks)

1. What is a manual information system?
2. What is CPU?
3. What is RAM?
4. What is a data bank?
5. Define TPS.
6. What is DSS?
7. Define decision making.
8. What is bounded rationality?
9. What is the difference between data and information?
10. What is batch processing?